

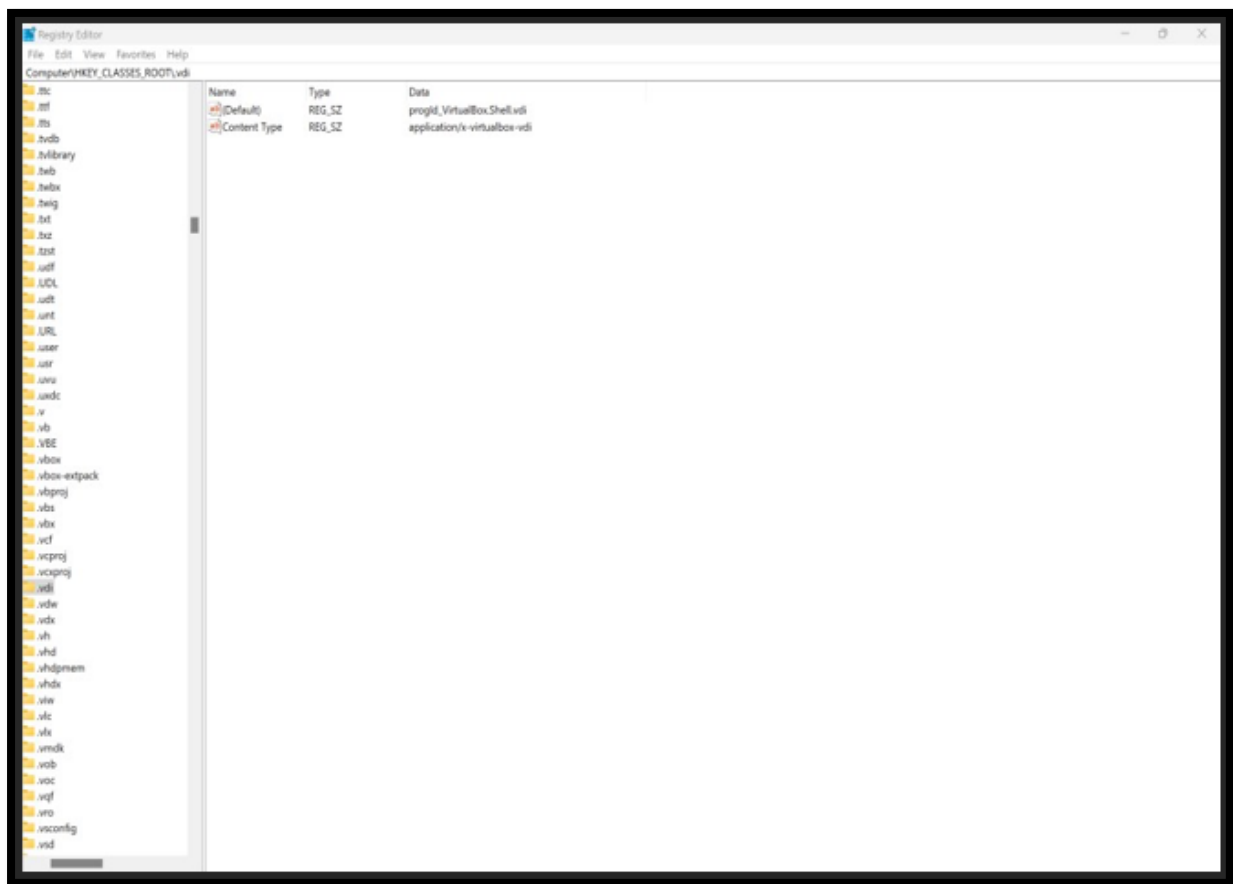
Exp – 7 Forensic Analysis Report , DFIR

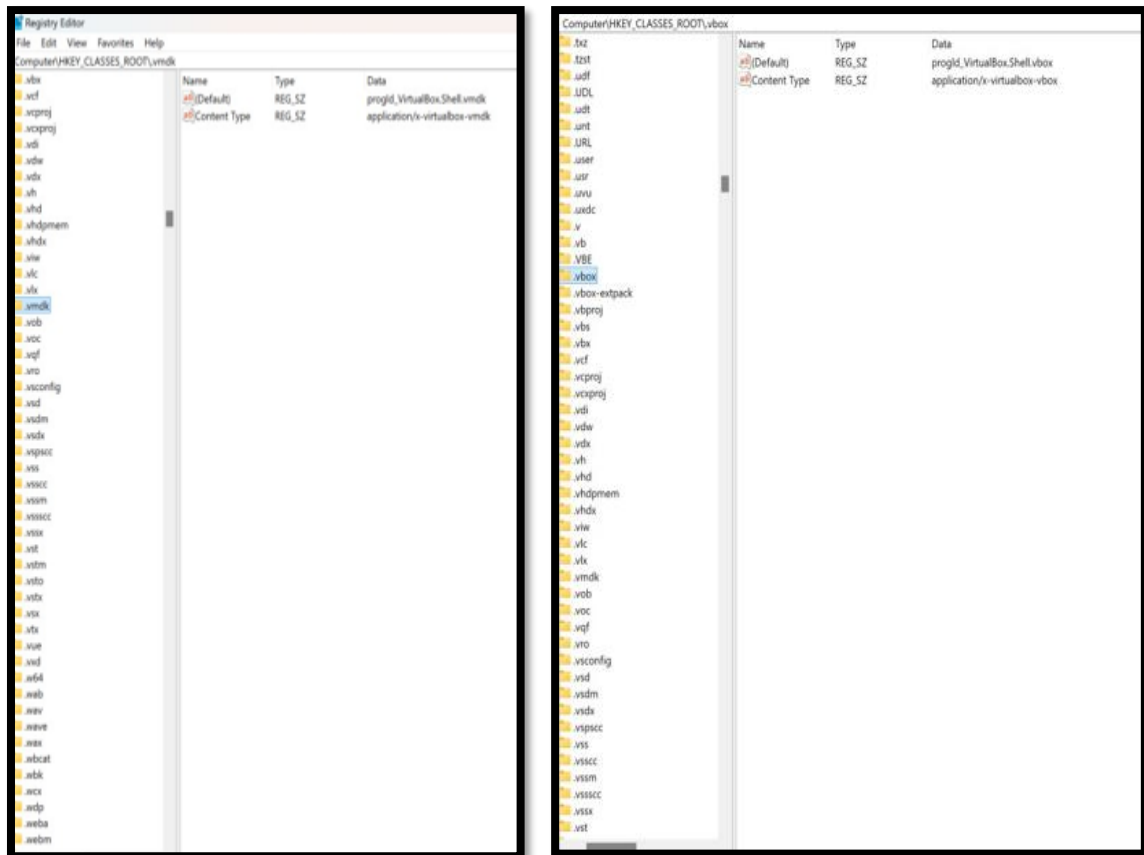
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1. Introduction

This report presents the findings of a forensic investigation conducted on a system suspected of hosting a virtual machine environment. The primary objectives of this investigation were to:

- Confirm the presence of a virtual machine environment by examining registry entries and file extensions (e.g., .vdi, .vmdk, .vbox), including indicators of Type 2 hypervisors.
- Acquire a forensic image of the identified virtual disk file using FTK Imager.
- Analyze the evidence tree from the acquired image to locate potential foreign security data or anomalies indicative of unauthorized or malicious activity.





- **Scope:**

The investigation includes the analysis of registry entries, the acquisition of the virtual disk image (VDI), and the review of the evidence tree in FTK Imager. Special attention is given to distinguishing between Type 1 (bare-metal) and Type 2 (hosted) hypervisors.

- **Objectives:**

1. **Determine Virtual Machine Presence:** Confirm via registry examination that a virtual machine environment is installed on the system.
2. **Identify Type 2 Hypervisors:** Look for indicators related to Type 2 hypervisors (e.g., VirtualBox, VMware Workstation/Player) that run on top of a host operating system.
3. **Acquire the Virtual Disk Image:** Use FTK Imager to image the identified virtual disk file.
4. **Examine the Evidence Tree:** Analyze the disk image's partition structure and directory tree for any foreign security data.

3. Methodology

3.1 Registry

Examination

- 1. Tools:** ○ Windows Registry Editor (regedit)
- 2. Procedure:**
 - **Virtualization Software Identification:**
 - Accessed the registry (e.g., HKEY_LOCAL_MACHINE\SOFTWARE) to locate references to virtualization software.
 - Searched for keywords and file extensions such as .vdi, .vmdk, and .vbox.
 - **Type 2 Hypervisor Indicators:**
 - Specifically examined registry keys associated with popular Type 2 hypervisors (e.g., VirtualBox, VMware Workstation/Player).
 - Verified that the presence of these keys and file references implies that the virtual machine environment is hosted on a conventional operating system rather than directly on hardware.
- 3. Findings:**
 - The registry contained multiple references to virtualization software, including entries that suggest the presence of a Type 2 hypervisor (e.g., VirtualBox keys and related file extensions).
 - These findings supported further investigation by confirming that a virtual machine exists on the system.

3.2 Acquisition of the Virtual Disk Image

- 1. Tool:** ○ FTK Imager
 - 2. Procedure:**
 - Launched FTK Imager on a forensically sound workstation.
 - Selected File > Create Disk Image and chose “Image File” as the source type.
 - Navigated to the location of the identified .vdi file, as determined from the registry and file system search.
 - Selected an appropriate image format (e.g., E01 or RAW) and designated a secure destination for the forensic image.
 - Recorded all settings, file paths, and generated hash values (e.g., MD5/SHA-1) for chain-of-custody purposes.
 - 3. Findings:**
 - The virtual disk image was successfully acquired with hash values verifying its integrity.
 - All acquisition steps were documented in accordance with forensic best practices.
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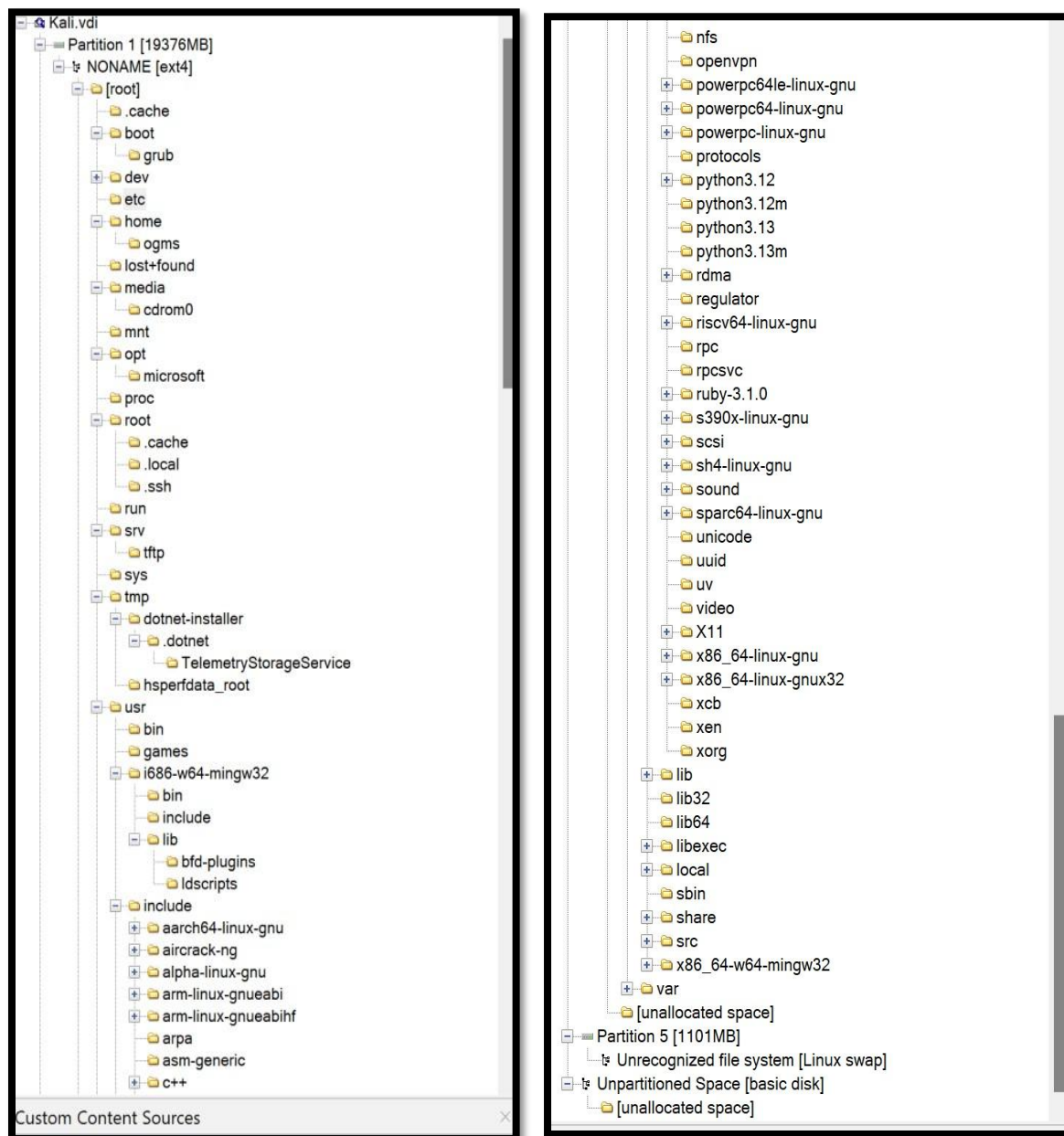
3.3 Evidence Tree Analysis

1. Tool: o FTK Imager

2. Procedure:

- o Loaded the acquired .vdi image in FTK Imager.
- o Reviewed the evidence tree to examine the partition structure and directory layout.
- o Focused on identifying partitions such as the root file system, Linux swap partitions, and unallocated space.

3. Exhibit A: Evidence Tree Screenshot



nfs
openvpn openp4le-linux-gnu modules
...
python2.7 python3.12 python3.9
...
xen x86
...

Partition 5 [1101MB] Unrecognized file system (Linux swap)
Unpartitioned Space [basic disk]
[unallocated space]

4. Observations:

- **Root Directories & Modules:**
The evidence tree shows numerous Linux-based directories (e.g., nfs, openvpn, python, xen), suggesting a Linux-based operating system or a hypervisor environment.
- **Swap Partition:**
Partition 5 is identified as a Linux swap partition (~1101MB). This area could contain remnants of in-memory data, such as credentials or logs.
- **Unallocated Space:**
Unallocated space indicates possible areas where deleted or hidden files may reside, warranting further forensic carving.

3.4 Identification of Foreign Security Data

1. Procedure:

- Searched for files and directories commonly associated with virtualization and security, such as configuration files (e.g., .vmx, .vbox), log files, and network configuration data.
- Analyzed the unallocated space and swap partition for residual data that may indicate unauthorized or malicious activity.

2. Potential Indicators:

- **Networking Tools:**
The presence of directories like openvpn may point to remote access configurations, which should be cross-examined for legitimacy.
- **Configuration Files:**
Files tied to the Type 2 hypervisors (e.g., VirtualBox configuration files) were examined for unauthorized modifications.
- **Memory Artifacts:**
The Linux swap partition may contain fragments of in-memory data, potentially revealing sensitive or security-related information.
- **Unallocated Space Artifacts:**
Data remnants in the unallocated space were noted as possible repositories of hidden or deleted malicious files.

3. Preliminary Findings:

- The directory structure and partition layout suggest a complex virtual environment with both authorized virtualization components and areas that might harbor foreign security data.
- Further detailed analysis is necessary to determine if any malicious modifications or unauthorized data exfiltration has occurred.

4. Results & Interpretation

- **Virtual Environment Confirmation:**

Registry examination and file system analysis confirm that the system hosts a virtual machine, including evidence of a Type 2 hypervisor (e.g., VirtualBox or VMware Workstation/Player).

- **Successful Image Acquisition:**

The forensic image of the .vdi file was successfully created using FTK Imager, with documented hash values ensuring data integrity.

- **Evidence Tree Insights:**

The evidence tree revealed a Linux-based environment with multiple directories (e.g., openvpn, python, xen) and notable partitions (Linux swap and unallocated space) that may contain critical forensic artifacts.

- **Foreign Security Data Considerations:**

The presence of advanced networking tools and configuration files, along with unexplored areas such as unallocated space and swap partitions, suggest that additional detailed analysis is warranted to rule out unauthorized or malicious activity.

5. Conclusions & Recommendations

Conclusions:

- The system clearly hosts a virtual machine environment, as confirmed by registry entries, file extensions, and the presence of a .vdi file.
- Evidence of a Type 2 hypervisor was found, indicating that the virtual environment is hosted on a conventional operating system rather than running directly on hardware.
- The forensic acquisition of the virtual disk image was successfully performed, and initial evidence tree analysis has identified key areas (e.g., swap partition, unallocated space) for further investigation.
- Preliminary findings suggest that foreign security data may be present; however, additional in-depth analysis of log files, configurations, and memory artifacts is required.

Recommendations:

- 1. In-Depth Analysis of Log & Configuration Files:**

- Review system logs (e.g., located in /var/log/ or hypervisor-specific directories) for anomalies.
- Verify configuration file integrity, especially those related to virtualization (e.g., .vbox or .vmx files).

- 2. Carving and Memory Analysis:**

- Perform forensic carving on the unallocated space to recover deleted or hidden files.

- Conduct a detailed analysis of the swap partition for memory-resident artifacts, which might include sensitive data or traces of unauthorized activity.
- 3. Cross-Referencing Data:**
- Compare findings with network logs and other security data sources to ascertain the legitimacy of remote access or configuration changes.
 - Ensure continuous documentation and chain-of-custody compliance throughout the investigation.
- 4. Preservation and Further Investigation:**
- Securely preserve the acquired image and all related forensic data.
 - If foreign security data is confirmed, initiate additional forensic or legal proceedings as appropriate.

6. Attachments

- **Exhibit A: Evidence Tree Screenshot (as provided)**
 - **Exhibit B: Hash Values and Acquisition Logs from FTK Imager**
 - **Exhibit C: Registry Key Paths / Screenshots Indicating Virtualization Software and Type 2 Hypervisor Indicators**
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